

ORBITGUARD

Space Debris Tracking, Astrodynamic Conjunction Assessment & AI Mission Decision Support

Problem Statement ID:	PS-04
Problem Statement Title:	Space Debris Tracking & Satellite Collision Risk Prediction Dashboard
Theme:	Space Technology / Defense & Security / Advanced Software
PS Category:	Software (Full-Stack Astrodynamics & AI Mission Platform)
Team ID / Team Name:	[Team ID] • [Team Name Registered on SIH Portal]
Platform Status:	ORBITGUARD v2.1.0-PROD • 100% Validated • 79/79 Tests Passing

PROPOSED SOLUTION (Describe your Idea / Solution / Prototype)

1. Multi-Source Ingestion CelesTrak & Space-Track live feeds + Modulo-10 checksum validation with 35,535 TLE cache.	2. SGP4 Astrodynamics SGP4 propagation, 3-tier spatial sieve (altitude shells <=50km), and orthogonal Secant TCA solver.
3. Probability & Risk Foster-2D B-plane numerical integral, Akella-Alfriend series, Alfano max Pc, and 0-100 risk scoring.	4. CAM & AI Decision 4-candidate impulsive burn matrix, Tsiolkovsky Hydrazine fuel cost, and 15-tool grounded AI copilot.

End-to-End Operational Workflow:

DATA	Multi-source live TLE ingestion (CelesTrak / Space-Track) + Modulo-10 checksums.
SGP4	Vectorized SGP4 propagation in TEME frame with WGS-84 / EGM-96 geoid transforms.
TCA	3-tier spatial sieve + Secant root solver finding exact orthogonal TCA ($r_{rel} \cdot v_{rel} = 0$).
Pc & RISK	Foster-2D B-plane probability integral + Alfano max Pc + 0-100 composite aerospace risk score.
CAM	Gauss variational equations compute 4 burn candidates + Tsiolkovsky Hydrazine propellant mass.
DECISION	FDO / Flight Director approval controls, post-CAM verification, and immutable UTC audit logging.

TECHNICAL APPROACH

3-Layer Architecture & Implementation Methodology

LAYER 1: DATA & FEED INGESTION • Feeds: Celestrak GP API (Primary), Space-Track 18th SDS, SatNOGS. • Ingestion: Async HTTP fetcher + Modulo-10 line checksum validation. • Fallback: Local verified cache with 35,535 real space objects. • Provenance: Real-time UI badge explicitly separating LIVE from DEMO.	LAYER 2: ASTRODYNAMICS ENGINE • Propagator: SGP4 C-extension (Vallado 2006) with J2/J3/J4 harmonics. • 3-Tier Sieve: Altitude shells ($\Delta h \leq 50\text{km}$) prune 98.6% pairs in $O(1)$. • Secant TCA: Orthogonal zero-crossing ($r_{\text{rel}} \cdot v_{\text{rel}} = 0$) in $<10^{-4}$ s. • Pc & Risk: Foster-2D B-plane integral + Akella-Alfriend + Alfano max Pc.	LAYER 3: DECISION & CLIENT • CAM: 4-candidate matrix + Tsiolkovsky fuel mass ($I_{\text{sp}}=220\text{s}$ Hydrazine). • AI Copilot: 15-tool layer + Digit Validator ($\pm 1\%$ tolerance). • Case Center: 13-section HUD (NEW -> APPROVED -> VERIFIED). • 3D WebGL: Three.js rendering 35k objects @ 60 FPS + CCSDS CDM.
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Verified Technology Stack: Python 3.9+ • FastAPI • SGP4 C-Extension • NumPy • SciPy • SQLAlchemy • SQLite • React 18 • TypeScript • Three.js WebGL • TailwindCSS

FEASIBILITY AND VIABILITY

Analysis, Operational Risks & Mitigation Strategies

CHALLENGE → OPERATIONAL RISK → MITIGATION	EMPIRICAL BENCHMARKS
1. TLE Covariance Gap: TLEs lack 6x6 covariance matrices. → <i>Mitigation:</i> Computes Alfano Max Pc bounds ($P_{c,max} = R^2 / e \cdot d^2$) establishing theoretical worst-case risk limits.	79 / 79 Tests Passed (100%) Automated Pytest test suite across unit & integration tests.
2. External Feed Outage: Space-Track / CelesTrak downtime. → <i>Mitigation:</i> Multi-source failover architecture with instant fallback to local verified cache (35,535 TLEs).	11 / 11 Benchmarks Passed (100%) Validated against Vallado 2006 standards in 0.0008s.
3. AI Hallucination Risk: LLM generating false numbers. → <i>Mitigation:</i> Zero-LLM physics rule + automated Digit Validator ($\pm 1\%$ tolerance) verifying all claims against physics engine.	~100.7 ms Pipeline Latency Catalog screening latency across 35,000+ objects.
4. Real Thruster Risk: Accidental unverified firing. → <i>Mitigation:</i> Framed as Decision Support. Real commanding requires multi-sig human approval & encryption keys.	+26.9 km Miss Clearance Delivered via 0.505 m/s impulsive prograde burn.
5. Computational Scaling: $O(N^2)$ all-pairs bottleneck. → <i>Mitigation:</i> 3-tier spatial sieve prunes 98.6% of candidate pairs in $O(1)$ time via altitude envelope shells.	117 grams Hydrazine Propellant consumed for 260 kg spacecraft (Tsiolkovsky).
	-99.9% Risk Reduction Achieved post-CAM verification clearance.

IMPACT AND BENEFITS

Social, Economic, Environmental & Operational Beneficiary Reach

Workflow Progression: Early Screening → Precise Risk → Faster Decision → Optimal CAM → Reduced Risk → Sustainability

■■ SATELLITE OPERATORS • Democratizes enterprise-grade CARA & CAM optimization without high recurring SaaS fees. • Exact fuel cost modeling (117g Hydrazine per CAM) with 99.9% risk reduction.	■■ SPACE AGENCIES & DEFENSE • Sovereign, explainable, and air-gapped conjunction monitoring with full data provenance. • Automated CCSDS 508.0-B-1 CDM export and Defense SITREP executive threat dossiers.
■ RESEARCH & ACADEMIA • Open-access, transparent astrodynamics testbed for training next-generation space engineers. • 100% mathematically proven physics equations with zero black-box heuristics.	■ SPACE SUSTAINABILITY • Directly mitigates Kessler Syndrome runaway cascade risk, safeguarding critical LEO/GEO orbits. • Protects multi-billion-dollar space assets from hypervelocity kinetic debris destruction.

RESEARCH, VALIDATION & REFERENCES

Analytical Proofs, Benchmark Evidence & International Standards

SCIENTIFIC VALIDATION BENCHMARKS ✓ SGP4 Propagation: Validated against Vallado 2006 Reference Vectors (LEO-00005, ISS-25544) with <0.001 km epoch error. ✓ Coordinate Transforms: TEME -> ECEF -> WGS-84 Geodetic validated using GMST rotation and Bowring's closed-form algorithm. ✓ Secant TCA Solver: Resolves exact orthogonal encounter condition ($r_{rel} \cdot v_{rel} = 0$) with sub-millisecond precision ($<10^{-4}$ s). ✓ Foster-2D Pc Engine: B-plane numerical integral & Akella-Alfriend series match NASA benchmarks (<0.1% deviation). ✓ Gauss CAM Mechanics: Along-track separation ($\Delta s \approx 3 \Delta v_t \Delta t_{lead}$) & Tsiolkovsky Hydrazine propellant mass validated. ✓ Test Pass Rate: 79/79 Pytest tests (100%) & 11/11 analytical benchmarks passing in 0.0008s.	RESEARCH PAPERS & INTERNATIONAL STANDARDS • SGP4 Standard: Vallado, D. A., et al. (2006). <i>Revisiting Spacetrack Report #3: Rev 2</i>. AIAA 2006-6753. • Foster-2D Collision Probability: Foster, J. L., & Estes, H. S. (1992). <i>A Parametric Analysis of Orbital Debris Collision Probability and Maneuver Utility</i>. NASA JSC-25898. • Short-Encounter Model: Chan, K. (1997). <i>Collision Probability for Space Missions</i>. Advances in Astronautical Sciences, 96. • Akella-Alfriend Probability: Akella, M. R., & Alfriend, K. T. (2000). <i>Probability of Collision Between Space Objects</i>. JGCD, 23(5). • Alfano Maximum Pc Bounds: Alfano, S. (2005). <i>Relating Position Uncertainty to Maximum Conjunction Probability</i>. JAS, 53(2). • CCSDS CDM Standard: CCSDS 508.0-B-1 (2013). <i>Conjunction Data Message (CDM)</i>. Blue Book Standard.
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